

functional_block.v

```
module functional_block (  
    input clk,  
    input rst,  
    input enable,  
    output reg [3:0] counter  
);  
always @(posedge clk or posedge rst) begin  
    if (rst)  
        counter <= 0;  
    else if (enable)  
        counter <= counter + 1;  
end  
endmodule
```